

EEE4114F

Backpropagation

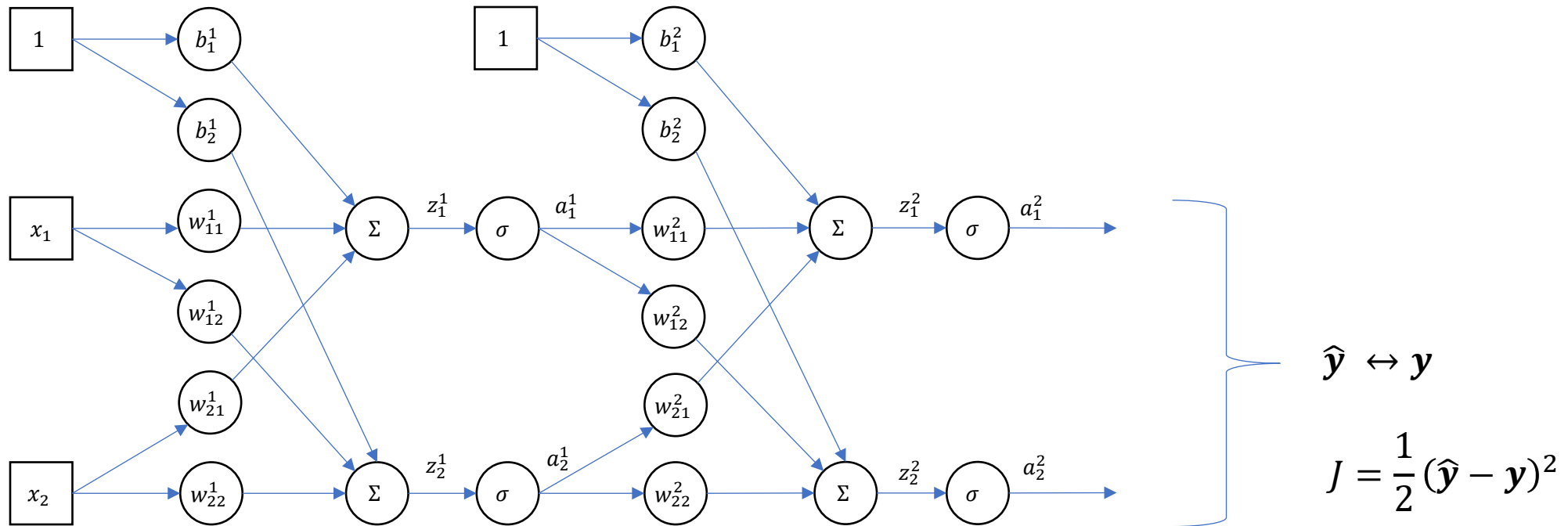


How do we learn weights?

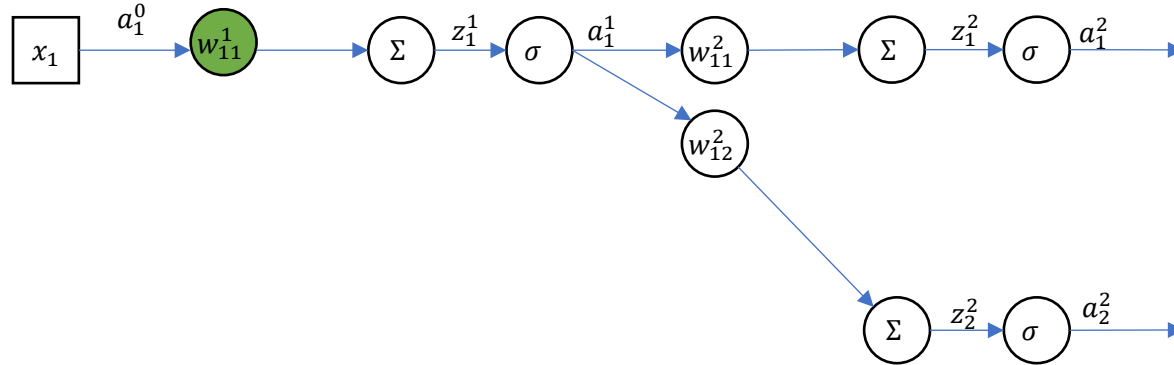
- We can use gradient descent
 - Easy for single layer e.g. Linear/logistic regression
 - Adding layers makes it more complicated to calculate gradients
- Backpropagation
 - Functions of functions => Chain rule
 - Calculate error from forward pass and determine gradients via chain rule



Backpropagation



Backward pass



$$\frac{\partial J}{\partial w_{11}^1} = \frac{\partial J}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial a_1^1} \cdot \frac{\partial a_1^1}{\partial z_1^1} \cdot \frac{\partial z_1^1}{\partial w_{11}^1} + \frac{\partial J}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial a_1^1} \cdot \frac{\partial a_1^1}{\partial z_1^1} \cdot \frac{\partial z_1^1}{\partial w_{11}^1}$$

What a nightmare!

$\frac{\partial J}{\partial a_1^2}$ easy to compute for last layer

$$\frac{\partial}{\partial a_1^2} \left[\frac{1}{2} (\hat{\mathbf{y}} - \mathbf{y})^2 \right] = \hat{y}_1 - y = a_1^2 - y$$

$$\frac{\partial a_1^2}{\partial z_1^2} = \sigma'(z_1^2)$$

$$z_1^2 = w_{11}^2 a_1^1 + w_{21}^2 a_2^1 + b_1^2$$

$$\frac{\partial z_1^2}{\partial a_1^1} = w_{11}^2$$

$$a_1^1 = \sigma(z_1^1)$$

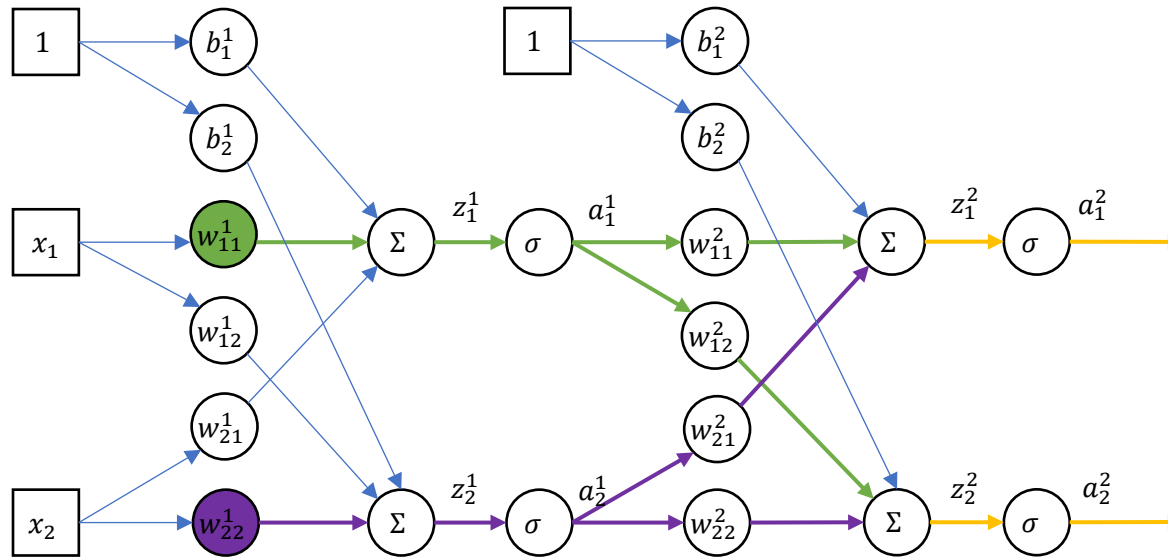
$$\frac{\partial a_1^1}{\partial z_1^1} = \sigma'(z_1^1)$$

$$z_1^1 = w_{11}^1 a_1^0 + w_{21}^1 a_2^0 + b_1^1$$

$$\frac{\partial z_1^1}{\partial w_{11}^1} = a_1^0 = x_1$$



Backward pass



$$\frac{\partial J}{\partial w_{11}^1} = \frac{\partial J}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial a_1^1} \cdot \frac{\partial a_1^1}{\partial z_1^1} \cdot \frac{\partial z_1^1}{\partial w_{11}^1} + \frac{\partial J}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial a_1^1} \cdot \frac{\partial a_1^1}{\partial z_1^1} \cdot \frac{\partial z_1^1}{\partial w_{11}^1}$$

$$\frac{\partial J}{\partial w_{22}^1} = \frac{\partial J}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} \cdot \frac{\partial z_1^2}{\partial a_2^1} \cdot \frac{\partial a_2^1}{\partial z_2^1} \cdot \frac{\partial z_2^1}{\partial w_{22}^1} + \frac{\partial J}{\partial a_2^2} \cdot \frac{\partial a_2^2}{\partial z_2^2} \cdot \frac{\partial z_2^2}{\partial a_2^1} \cdot \frac{\partial a_2^1}{\partial z_2^1} \cdot \frac{\partial z_2^1}{\partial w_{22}^1}$$

You might notice a few terms appear multiple times.

For convenience we can define the error in the k^{th} neuron in the l^{th} layer as

$$\frac{\partial J}{\partial z_k^l} = \delta_k^l$$

$$\frac{\partial J}{\partial a_1^2} \cdot \frac{\partial a_1^2}{\partial z_1^2} = \frac{\partial J}{\partial z_1^2} = \delta_1^2$$

$$\frac{\partial J}{\partial z_1^1} = \delta_1^1$$

$$= \delta_1^2 w_{11}^2 \sigma'(z_1^1) + \delta_2^2 w_{12}^2 \sigma'(z_1^1)$$

$$\frac{\partial J}{\partial w_{11}^1} = \delta_1^1 a_1^0 = \delta_1^1 x_1$$

Backpropagation equations – needs checking

Error in output layer:

$$\delta^L = \nabla_a J \odot \sigma'(\mathbf{z}^L)$$

Error in hidden layers:

$$\delta^l = \left((W^{l+1}) \delta^{l+1} \right) \odot \sigma'(\mathbf{z}^l)$$

Gradient of a given weight:

$$\frac{\partial J}{\partial w_{jk}^l} = \delta_k^l a_j^{l-1}$$

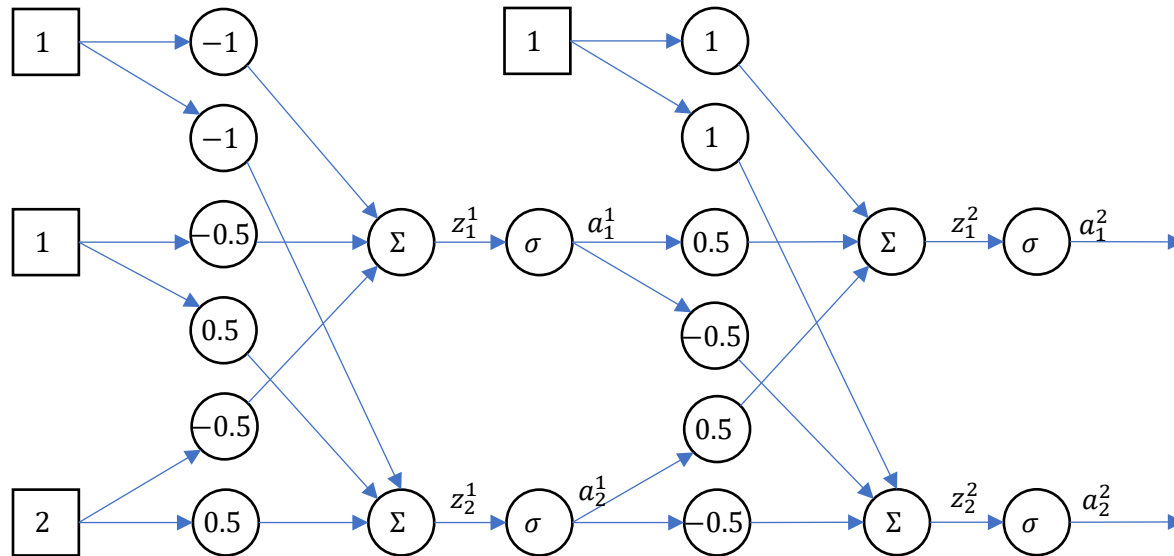
Gradient of a bias (same as above but with activation of 1):

$$\frac{\partial J}{\partial b_k^l} = \delta_k^l$$



Numerical Example - ReLU

$$\mathbf{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \mathbf{b}^1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, W^1 = \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}, W^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix}$$



$$\begin{aligned} \mathbf{z}^1 &= (W^1)^T \cdot \mathbf{x} + \mathbf{b}^1 \\ &= \begin{bmatrix} -0.5 & -0.5 \\ 0.5 & 0.5 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} -1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{a}^1 &= \sigma(\mathbf{z}^1) \\ &= \begin{bmatrix} \sigma(z_1^1) \\ \sigma(z_2^1) \end{bmatrix} = \begin{bmatrix} \max(0, z_1^1) \\ \max(0, z_2^1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{z}^2 &= (W^2)^T \cdot \mathbf{a}^1 + \mathbf{b}^2 \\ &= \begin{bmatrix} 0.5 & 0.5 \\ -0.5 & -0.5 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0.5 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 1.25 \\ 0.75 \end{bmatrix} \end{aligned}$$

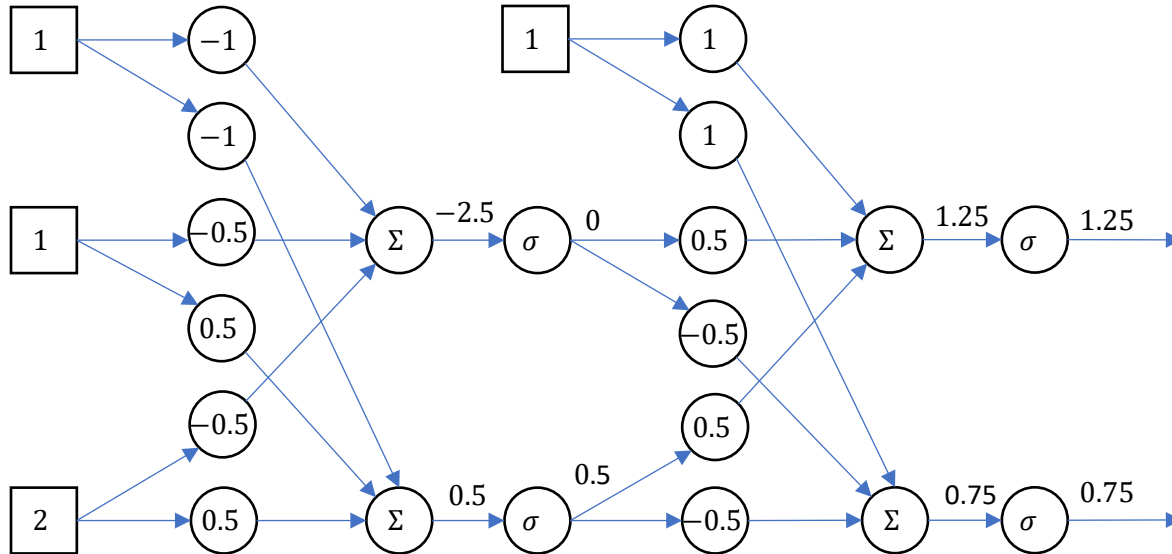
$$\begin{aligned} \mathbf{a}^2 &= \sigma(\mathbf{z}^2) \\ &= \begin{bmatrix} \sigma(z_1^2) \\ \sigma(z_2^2) \end{bmatrix} = \begin{bmatrix} \max(0, z_1^2) \\ \max(0, z_2^2) \end{bmatrix} = \begin{bmatrix} 1.25 \\ 0.75 \end{bmatrix} \end{aligned}$$



Calculate Error in Last Layer Neurons

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, J = \frac{1}{2} (\hat{\mathbf{y}} - \mathbf{y})^2$$

$$\boldsymbol{\delta}^L = \nabla_a J \odot \sigma'(\mathbf{z}^L)$$



$$\begin{aligned} \nabla_a J &= \begin{bmatrix} \frac{\partial J}{\partial a_1^2} \\ \frac{\partial J}{\partial a_2^2} \end{bmatrix} \\ &= \begin{bmatrix} a_1^2 - y_1 \\ a_2^2 - y_2 \end{bmatrix} = \begin{bmatrix} 1.25 - 0 \\ 0.75 - 1 \end{bmatrix} = \begin{bmatrix} 1.25 \\ -0.25 \end{bmatrix} \end{aligned}$$

$$\sigma'(\mathbf{z}^L) = \begin{bmatrix} ReLU'(z_1^2) \\ ReLU'(z_2^2) \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

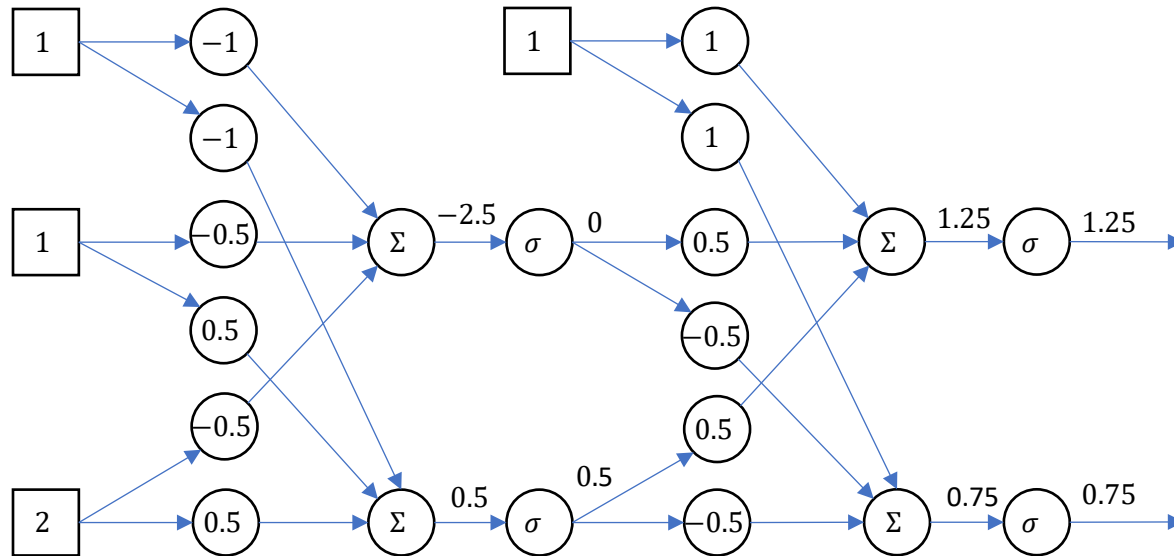
$$\boldsymbol{\delta}^2 = \begin{bmatrix} 1.25 \\ -0.25 \end{bmatrix} \odot \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1.25 \times 1 \\ -0.25 \times 1 \end{bmatrix} = \begin{bmatrix} 1.25 \\ -0.25 \end{bmatrix}$$



Calculate Error in Hidden Layer Neurons

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, J = \frac{1}{2} (\hat{\mathbf{y}} - \mathbf{y})^2$$

$$\delta^l = \left((W^{l+1}) \cdot \delta^{l+1} \right) \odot \sigma'(\mathbf{z}^l)$$



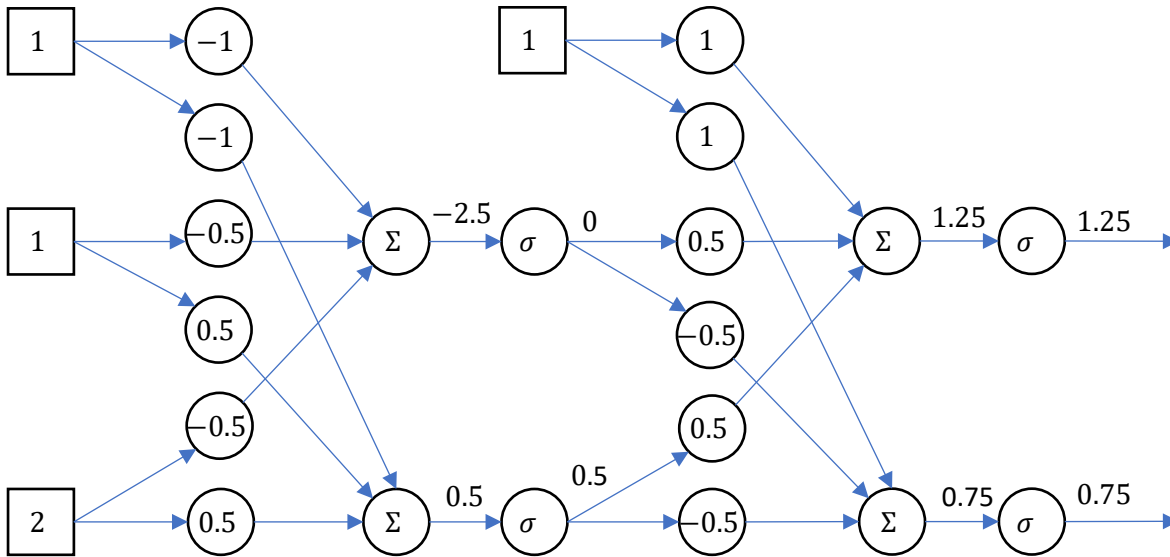
$$\begin{aligned} \delta^1 &= \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix} \cdot \begin{bmatrix} 1.25 \\ -0.25 \end{bmatrix} \odot \sigma' \left(\begin{bmatrix} -2.5 \\ 0.5 \end{bmatrix} \right) \\ &= \begin{bmatrix} 0.75 \\ 0.75 \end{bmatrix} \odot \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0.75 \end{bmatrix} \end{aligned}$$



Calculate Gradients for Model Parameters

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, J = \frac{1}{2} (\hat{\mathbf{y}} - \mathbf{y})^2$$

$$\frac{\partial J}{\partial W^l} = (\boldsymbol{\delta}^l)^T (\mathbf{a}^{l-1})$$



$$\frac{\partial J}{\partial W^2} = \begin{bmatrix} \frac{\partial J}{\partial w_{11}^2} & \frac{\partial J}{\partial w_{12}^2} \\ \frac{\partial J}{\partial w_{21}^2} & \frac{\partial J}{\partial w_{22}^2} \end{bmatrix} = (\boldsymbol{\delta}^2)^T \cdot (\mathbf{a}^1) = \begin{bmatrix} \delta_1^2 a_1^1 & \delta_1^2 a_2^1 \\ \delta_2^2 a_1^1 & \delta_2^2 a_2^1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1.25 \times 0.5 \\ 0 & -0.25 \times 0.5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0.625 \\ 0 & -0.125 \end{bmatrix}$$

$$\frac{\partial J}{\partial W^1} = \begin{bmatrix} \frac{\partial J}{\partial w_{11}^1} & \frac{\partial J}{\partial w_{12}^1} \\ \frac{\partial J}{\partial w_{21}^1} & \frac{\partial J}{\partial w_{22}^1} \end{bmatrix} = (\boldsymbol{\delta}^1)^T \cdot (\mathbf{a}^0) = \begin{bmatrix} \delta_1^1 a_1^0 & \delta_1^1 a_2^0 \\ \delta_2^1 a_1^0 & \delta_2^1 a_2^0 \end{bmatrix}$$

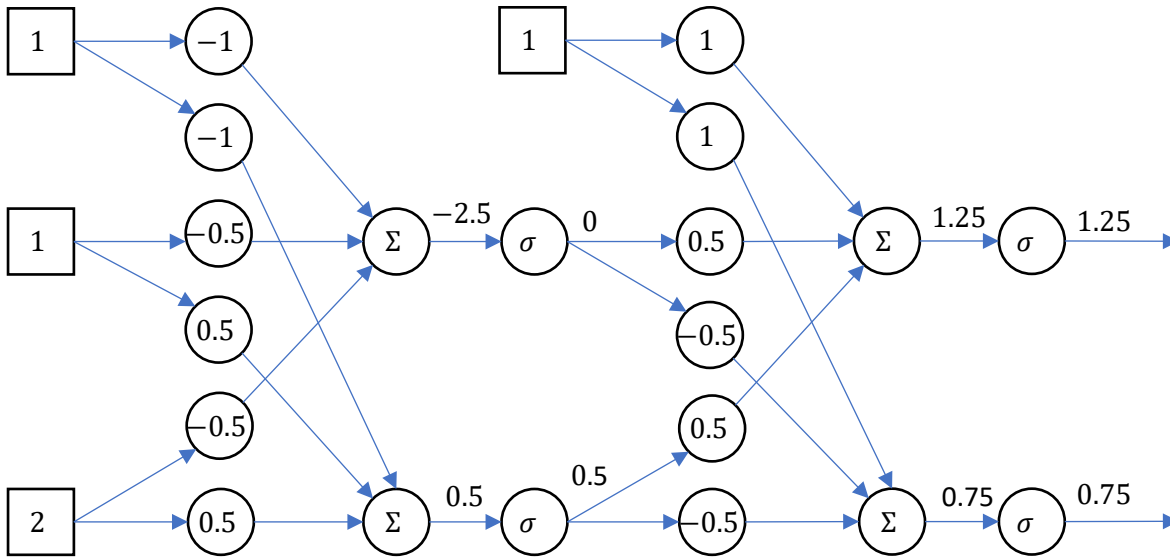
$$= \begin{bmatrix} 0 & 0 \\ 0.75 \times 1 & 0.75 \times 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0.75 & 1.5 \end{bmatrix}$$



Calculate Gradients for Model Parameters

$$\mathbf{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, J = \frac{1}{2}(\hat{\mathbf{y}} - \mathbf{y})^2$$



$$\frac{\partial J}{\partial \mathbf{b}^l} = \boldsymbol{\delta}^l$$

$$\begin{bmatrix} \frac{\partial J}{\partial b_1^2} \\ \frac{\partial J}{\partial b_2^2} \end{bmatrix} = \begin{bmatrix} \delta_1^2 \\ \delta_2^2 \end{bmatrix} = \begin{bmatrix} 1.25 \\ -0.25 \end{bmatrix}$$

$$\begin{bmatrix} \frac{\partial J}{\partial b_1^1} \\ \frac{\partial J}{\partial b_2^1} \end{bmatrix} = \begin{bmatrix} \delta_1^1 \\ \delta_2^1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0.75 \end{bmatrix}$$

Parameter Update

- Usual gradient descent rules

- $w := w - \eta \frac{\partial J}{\partial w}$

- $b := b - \eta \frac{\partial J}{\partial b}$

- $\mathbf{b}^1 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \mathbf{b}^2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, W^1 = \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix}, W^2 = \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix}$

- $\boldsymbol{\delta}^1 = \begin{bmatrix} 0 \\ 0.75 \end{bmatrix}, \boldsymbol{\delta}^2 = \begin{bmatrix} 1.25 \\ -0.25 \end{bmatrix}$

- $\frac{\partial J}{\partial \mathbf{b}^l} = \boldsymbol{\delta}^l$

- $\frac{\partial J}{\partial W^l} = (\boldsymbol{\delta}^l)^T (\mathbf{a}^{l-1})$



Parameter Update

Let $\eta = 0.1$

$$W^1 := \begin{bmatrix} -0.5 & 0.5 \\ -0.5 & 0.5 \end{bmatrix} - 0.1 \begin{bmatrix} 0 & 0 \\ 0.75 & 1.5 \end{bmatrix} = \begin{bmatrix} -0.5 & 0.5 \\ -0.575 & 0.515 \end{bmatrix}$$
$$W^2 := \begin{bmatrix} 0.5 & -0.5 \\ 0.5 & -0.5 \end{bmatrix} - 0.1 \begin{bmatrix} 0 & 0.625 \\ 0 & -0.125 \end{bmatrix} = \begin{bmatrix} 0.5 & -0.5625 \\ 0.5 & -0.4875 \end{bmatrix}$$

$$\mathbf{b}^1 := \begin{bmatrix} -1 \\ -1 \end{bmatrix} - 0.1 \begin{bmatrix} 0 \\ 0.75 \end{bmatrix} = \begin{bmatrix} -1 \\ -1.075 \end{bmatrix}$$
$$\mathbf{b}^2 := \begin{bmatrix} 1 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 1.25 \\ -0.25 \end{bmatrix} = \begin{bmatrix} 0.875 \\ 1.025 \end{bmatrix}$$



Next up:

Backpropagation continued

